# AyuAware My Twin Section

## Technical Product Requirements Document

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## 1. Executive Summary

### What We're Building

The My Twin section is the user's **relationship dashboard** — a place to understand how their Digital Twin sees them, how the Twin is learning and evolving, and how to customize the relationship. It answers the fundamental question: "*What does my Twin know about me?*"

This section combines two interaction paradigms:

1. **Conversational Introduction** — The Twin explains information before showing it
2. **Information Cards** — Dense, scannable, expandable data displays

The result is a hybrid experience where the Twin feels like a guide showing you around a museum of yourself, rather than a static profile page.

### Why It Matters

Trust is the foundation of the AyuAware experience. Users need to understand:

- **What the Twin knows** — Their constitution, patterns, and predictions
- **How it learned this** — Data sources, accuracy, confidence levels
- **How accurate it is** — Prediction track record, learning progress
- **How to control it** — Customization, data management, privacy

Without transparency, the Twin feels like a black box. With it, the Twin becomes a trusted partner whose insights carry weight.

### Key Differentiator

Most health apps have static profile pages that users visit once and forget. The My Twin section is **dynamic and evolving**:

- Learning progress updates weekly
- New patterns appear as they're discovered
- Predictions change daily based on context
- The Twin's accuracy improves over time

Users return not to update their profile, but to see how their Twin is growing alongside them.

# 2. Section Overview

## 2.1 The Hybrid Chat + Card Pattern

This section pioneered a unique pattern that we call "Conversational Exhibits":

```
┌──────────────────────────────────────────────────────────────────────────────────────────────────────────────┐
│                                                                                                              │
│                                                                                                              │
│                                                                                                              │
```

| Approach | Problem |
|---|---|
| Pure chat | Information is hard to scan, data gets lost in conversation |
| Pure cards/ dashboard | Feels cold, lacks personality, no explanation |
| Hybrid | Twin explains context, cards display data efficiently |

## 2.2 Information Architecture

The section organizes around five major themes:

```
My Twin Section
│
├── 📊 Learning Progress
│   └── How the Twin is evolving, accuracy metrics
│
├── 🧬 Constitution
│   ├── Prakriti (birth constitution)
│   └── Vikriti (current state)
│
├── 🎓 Intelligence Evolution
│   └── Capability levels, unlocking features
│
├── 🔍 Patterns Discovered
│   └── Correlations found in user's data
│
├── 🔮 Predictions
│   └── Upcoming forecasts with prevention tips
│
└── ⚙️ Customization
    └── Communication style, notifications, privacy
```

## 2.3 Design Philosophy

**Twin Speaks First** Every piece of information is introduced conversationally. Instead of a static "Constitution" card appearing, the Twin says:

"Your constitution is the foundation of how I understand you. Let me show you what I see:"

**Cards as Exhibits** Information cards feel like exhibits in a museum — the Twin is your guide, walking you through each display and explaining what you're looking at.

**Progressive Disclosure**

- **Collapsed**: Key stats visible at a glance (3 seconds to scan)
- **Expanded**: Full detail available with one tap
- **Deep Dive**: Links to dedicated screens for comprehensive exploration

**Transparency Builds Trust**

- Show confidence percentages on predictions
- Display observation counts behind patterns
- Reveal the learning journey with milestones
- Explain *why* predictions are being made

# 3. User Journey Breakdown

## 3.1 First Visit to My Twin

**Context:** User has been using the app for a few weeks and navigates to My Twin for the first time.

**What They See:**

The Twin greets them and immediately surfaces the most relevant information:

**Twin Message 1:**

"Hi Sarah! I'm your Digital Twin.

We've been learning together for 45 days. Want to see how I'm understanding you?"

**Learning Progress Card (Collapsed):**

```
┌──────────────────────────────────────────────┐
│ 📊 My Learning Progress                      │
│ ──────────────────────────────────────────── │
│                                              │
│ 🧠 Advanced Intelligence                     │
│ 📈 87% Accuracy (↑ from 65%)                 │
│                                              │
│ • 45 days active                             │
│ • 847 data points analyzed                   │
│ • 127 patterns discovered                    │
│                                              │
│            [View Details ↓]                  │
└──────────────────────────────────────────────┘
```

**Twin Message 2:**

"What would you like to explore?"

🧬 Your constitution 📈 How I'm learning 🎯 Patterns I've found 🔮 My predictions ⚙️ Customize me

**Why This Works:**

- Personal greeting establishes relationship
- Key stats visible immediately (no hunting)

- User chooses their own path
- Feels like a conversation, not a profile page

## 3.2 Exploring Constitution

**Context:** User taps "🧬 Your constitution" from the menu.

**Twin Message:**

"Your constitution is the foundation of how I understand you. Let me show you what I see:"

**Constitution Card (Collapsed):**

```
┌──────────────────────────────────────────┐
│ 🧬 Your Constitution                      │
│ ──────────────────────────────────────────│
│                                           │
│ [Toggle: ⚪ Prakriti  🔴 Vikriti]          │
│                                           │
│        [Dosha Circular Chart]             │
│           Primary: Pitta                  │
│                                           │
│ Current State (Vikriti):                  │
│ Pitta 45% • Vata 40% • Kapha 15%          │
│                                           │
│ ⚠️ Vata increased 15% from baseline       │
│                                           │
│           [See More Details ↓]            │
└──────────────────────────────────────────┘
```

**User Taps "See More Details":**

The card expands to reveal:

1. **Birth Constitution (Prakriti)** — With assessment date
2. **Current State (Vikriti)** — With last update
3. **Imbalance Alert** — Explaining the Vata increase
   - Symptoms this may be causing
   - Contributing factors the Twin has noticed
4. **Physical Traits** — What Twin knows about their body
5. **Mental Qualities** — Cognitive and emotional patterns
6. **Tendencies** — Vulnerabilities and strengths
7. **Action Buttons:**
   - [View Balancing Protocol →]
   - [See Constitution History →]

**Why This Works:**

- Toggle lets user compare ideal vs. current state
- Alert immediately surfaces actionable insight

- Explanation connects dots they might not see
- Actions available right where they're relevant

### 3.3 Understanding Intelligence Evolution

**Context:** User wants to understand how the Twin's capabilities grow.

**Twin Message:**

"I'm constantly evolving as I learn more about you. Here's my current intelligence level and what's next:"

**Intelligence Evolution Card (Collapsed):**

```
┌──────────────────────────────────────────────┐
│ 🎓 Intelligence Evolution                    │
│ ──────────────────────────────────────────── │
│                                              │
│ Current Level: Week 8                        │
│ 🔵 Short-term Prediction                     │
│                                              │
│ ━━━━━━━━━━━━━○○○○○○○○○○  68%                 │
│                                              │
│ Forecasting 1-3 days ahead                   │
│ 523 data points to next level                │
│                                              │
│           [View All Levels ↓]                │
└──────────────────────────────────────────────┘
```

**User Taps "View All Levels":**

Expanded card shows the full capability ladder:

| Level | Name | Status | Description |
|---|---|---|---|
| 1 | Basic Analysis | ✅ Completed | Constitutional assessment, simple patterns |
| 2 | Pattern Recognition | ✅ Completed | Lifestyle-symptom correlations |
| 3 | Short-term Prediction | 🔵 Current (68%) | Forecasting 1-3 days ahead |
| 4 | Long-term Forecasting | 🔒 Locked | 2-week predictions with seasonal adjustment |
| 5 | Genomic Integration | 🔒 Locked | DNA-level personalization |

**Also Shows:**

- How data points contribute to learning
- Estimated time to next level unlock
- User's contribution breakdown (check-ins, syncs, etc.)

**Why This Works:**

- Gamification creates investment in daily engagement
- Transparency shows the Twin isn't magic — it learns
- Progress is tangible and rewarding
- Unlocking features gives long-term motivation

## 3.4 Discovering Patterns

**Context:** User taps "🎯 Patterns I've found" to see what the Twin has learned.

**Twin Message:**

"I've been studying your patterns! Here are the most significant ones I've discovered this week:"

**Patterns Card (Collapsed):**

```
┌──────────────────────────────────────────────────┐
│ 🔍 Patterns Discovered                           │
│ ────────────────────────────────────────────────│
│                                                  │
│ 127 patterns learned from your data              │
│                                                  │
│ Top Insights This Week:                          │
│                                                  │
│ ☕ Coffee after 2pm → -20% sleep quality   92%   │
│                                                  │
│ 🍽️ Heavy lunch → Energy crash             88%   │
│                                                  │
│ 🌕 Full moon → Pitta spike                 85%   │
│                                                  │
│           [See All Patterns ↓]                   │
└──────────────────────────────────────────────────┘
```

**User Taps "See All Patterns":**

Each pattern expands to show:

- **Observation count** — "Observed 23 times over 45 days"
- **Category** — Sleep, Energy, Mood, Digestion, etc.
- **Detailed explanation** — What happens and why
- **Actions:**
  - [Create Protocol] — Turn pattern into actionable guidance
  - [See Timeline] — View when this pattern occurred

**Additional Patterns Listed:**

- Morning yoga → +25% day energy (91%)
- Screen time after 10pm → Restless (89%)
- Spicy dinner → Heartburn (87%)

- Meditation → -30% stress (86%)

**Why This Works:**

- Concrete evidence of Twin's learning
- Confidence percentages build trust
- Patterns are immediately actionable
- User can explore the data behind insights

## 3.5 Reviewing Predictions

**Context:** User wants to see what the Twin is forecasting.

**Twin Message:**

"Based on your recent patterns, I'm forecasting some things you should watch out for:"

**Predictions Card (Collapsed):**

```
┌──────────────────────────────────────────────┐
│ 🔮 Upcoming Predictions (3)                  │
│ ──────────────────────────────────────────── │
│                                              │
│ ⚠️ Tomorrow, 2pm              87%            │
│ Digestive discomfort likely                  │
│                                              │
│ ⚠️ In 3 days                  78%            │
│ Stress spike likely                          │
│                                              │
│ ⚠️ In 5 days                  65%            │
│ Sleep disruption possible                    │
│                                              │
│ 💡 Prediction accuracy: 87% (↑ from 65%)     │
│                                              │
│           [See Full Details ↓]               │
└──────────────────────────────────────────────┘
```

**User Taps "See Full Details":**

Each prediction expands to show:

**Prediction Detail (Tomorrow's Digestive Discomfort):**

```
⚠️ Tomorrow, 2pm                                [87%]

Digestive discomfort likely

Why I'm predicting this:
• Full moon phase (Pitta aggravating)
```

```
• Hot weather forecast (28°C)
• Your spicy lunch tendency on Fridays

Prevention tips:
• Skip the spicy food tomorrow
• Have cooling lunch (salad, mint tea)
• Take a short walk after eating

[Create Prevention Protocol]
```

**Also Shows:**

- Accuracy tracking with recent prediction history
- ✅/❌ indicators for past predictions
- Link to full prediction history

**Why This Works:**

- Predictions feel like looking into the future
- Transparency on *why* builds trust
- Prevention tips make predictions actionable
- Track record shows Twin is reliable

## 3.6 Customizing the Twin

**Context:** User wants to adjust how the Twin communicates.

**Twin Message:**

"I can adapt to your preferences! Let's customize how we communicate:"

**Customization Card:**

```
┌──────────────────────────────────────────┐
│ ⚙️ Customize Your Twin                    │
│ ──────────────────────────────────────── │
│                                          │
│ 💬 Communication Style                    │
│ ──────────────────────────────────────── │
│                                          │
│ Personality:                             │
│ ○ Supportive Friend                      │
│   "Hey! Great sleep last night! 🎉"      │
│                                          │
│ ● Wise Teacher (Current)                 │
│   "Your sleep improved due to..."        │
│                                          │
│ ○ Clinical Guide                         │
│   "Sleep efficiency: 92%. HRV: +15%"     │
```

```
│                                                          │
│ ──────────────────────────────────────────────────────── │
│                                                          │
│ Detail Level:                                            │
│ ○ Brief   ● Balanced   ○ Detailed                        │
│                                                          │
│ Sanskrit Usage:                                          │
│ ● Minimal   ○ Moderate   ○ Extensive                     │
│                                                          │
│            [More Settings ↓]                             │
└──────────────────────────────────────────────────────────┘
```

**Expanded Settings Include:**

**Voice & Language:**

- Voice personality (Warm, Professional, Energetic)
- Speech speed slider

**Notifications:**

- Morning summary time
- Prediction alerts toggle
- Check-in reminders toggle
- Do Not Disturb hours

**Data & Privacy:**

- Connected sources (Apple Health, Oura, etc.)
- Practitioner sharing preferences
- Links to manage data and view privacy policy

**Why This Works:**

- User feels in control of the relationship
- Examples show what each style sounds like
- Privacy settings build trust
- Changes take effect immediately

# 4. Technical Architecture

## 4.1 High-Level Architecture

```
┌──────────────────────────────────────────────────────────
```

```
│   │  ┌──────────┐ ┌──────────────────┐ ┌──────────────────┐
│   │  │ Header   │ │  Chat + Cards    │ │ Navigation       │
│   │  │ Bar      │ │  Hybrid View     │ │ Menu             │
│   │  └──────────┘ └──────────────────┘ └──────────────────┘
│
└──────────────────────────────────────────────────────────────
─┘   │
└──────────────────────────────────────────────────────────────
──────────┘
                                  │
                                  ▼
┌──────────────────────────────────────────────────────────────
──────────┐
│                          STATE LAYER
│
│   ┌──────────────────┐   ┌──────────────────┐
┌──────────────────┐
│   │ Twin Profile     │   │ Learning Data    │   │ Preferences
│   │ Store            │   │ Store            │   │ Store
│   │                  │   │                  │   │
│   │ • Constitution   │   │ • Progress       │   │ • Comm style
│   │ • Patterns       │   │ • Accuracy       │   │ •
Notifications  │   │
│   │ • Predictions    │   │ • Data points    │   │ • Privacy
│   │ • History        │   │ • Milestones     │   │ •
Connections    │   │
│   └──────────────────┘   └──────────────────┘
└──────────────────┘   │
└──────────────────────────────────────────────────────────────
──────────┘
                                  │
                                  ▼
┌──────────────────────────────────────────────────────────────
──────────┐
│                          SERVICE LAYER
│
```

```
│   ┌──────────────────────┐                                  │
┌──────────────────────────────────────────┐    │
│   │    My Twin Agent     │  │      Profile Service
│   │                      │  │
│   │    (Claude API)      │  │
│   │                      │  │
│   │                      │  │  • Fetch constitution data
│   │                      │  │
│   │  • Introductions     │  │  • Calculate patterns
│   │                      │  │
│   │  • Explanations      │  │  • Generate predictions
│   │                      │  │
│   │  • Responses         │  │  • Update preferences
│   │                      │  │
│   └──────────────────────┘  │
└──────────────────────────────────────────┘    │
└──────────────────────────────────────────────────────────────────────────────────────────┘
                                                  │
                                                  ▼
┌──────────────────────────────────────────────────────────────────────────────────────────┐
│                                   DATA LAYER
│
│   ┌──────────────────────┐                                  │
┌──────────────────────────────────────────┐    │
│   │   Local Cache        │  │      Backend APIs
│   │                      │  │
│   │                      │  │
│   │  • Profile data      │  │  • Constitution service
│   │                      │  │
│   │  • Pattern history   │  │  • Pattern analysis service
│   │                      │  │
│   │  • Preferences       │  │  • Prediction engine
│   │                      │  │
│   │  • UI state          │  │  • User preferences service
│   │                      │  │
│   └──────────────────────┘  │
└──────────────────────────────────────────┘    │
└──────────────────────────────────────────────────────────────────────────────────────────┘
```

### 4.2 Data Flow: Opening My Twin Section

```
User navigates to My Twin
```

```
        │
        ▼
┌──────────────────────────┐
│ Load cached profile      │ ── Instant display of last-known
data
└──────────────────────────┘
        │
        ▼
┌──────────────────────────┐
│ Generate greeting        │ ── Claude creates personalized
welcome
│ message                  │    based on days active, recent
activity
└──────────────────────────┘
        │
        ▼
                             Parallel requests:
┌──────────────────────────┐
│ Fetch fresh data         │ ── • GET /profile/constitution
└──────────────────────────┘    • GET /profile/learning-
progress
        │                    • GET /profile/patterns
        │                    • GET /profile/predictions
        ▼
┌──────────────────────────┐
│ Build Learning           │ ── Create collapsed card with
│ Progress card            │    latest stats
└──────────────────────────┘
        │
        ▼
┌──────────────────────────┐
│ Show navigation menu     │ ── Twin offers exploration options
└──────────────────────────┘
```

### 4.3 Data Flow: Card Expansion

```
User taps "View Details ↓"
        │
        ▼
┌──────────────────────────┐
│ Check if data cached     │
└──────────────────────────┘
        │
    ┌───┴───┐
    │       │
Cached    Not Cached
    │       │
    │       ▼
```

```
│
│      ┌──────────────────────────┐
│      │ Show loading state       │
│      │ in expansion area        │
│      └──────────────────────────┘
│            │
│            ▼
│      ┌──────────────────────────┐
│      │ Fetch detailed data      │
│      └──────────────────────────┘
└──────────┬
           ▼
┌──────────────────────────┐
│ Animate card             │ ── Height: 400ms ease-out
│ expansion                │    Content fade: 300ms delay
└──────────────────────────┘
           ▼
┌──────────────────────────┐
│ Render expanded          │
│ content                  │
└──────────────────────────┘
```

### 4.4 Data Flow: Customization Change

```
User changes communication style
           ▼
┌──────────────────────────┐
│ Optimistic update        │ ── Immediately reflect change in
UI
└──────────────────────────┘
           ▼
┌──────────────────────────┐
│ Persist locally          │ ── Save to AsyncStorage
└──────────────────────────┘
           ▼
┌──────────────────────────┐
│ Sync to backend          │ ── POST /preferences/update
└──────────────────────────┘
           ▼
┌──────────────────────────┐
│ Show confirmation        │ ── Toast: "Preferences saved"
```

```
└──────────────────────────┘
            │
            ▼
┌──────────────────────────┐
│ Apply to Claude          │ ── Update system prompt context
│ system prompt            │    for future messages
└──────────────────────────┘
```

# 5. AI Integration & Prompting Strategy

## 5.1 The Role of Claude in My Twin Section

Claude serves as the **voice and personality** of the Twin in this section. Its responsibilities:

| Capability | Description |
|---|---|
| Personalized Greetings | Welcome based on relationship duration and recent activity |
| Card Introductions | Explain what each card shows before displaying it |
| Insight Explanations | Describe patterns and predictions in accessible terms |
| Imbalance Analysis | Explain constitution changes and their implications |
| Response to Questions | Answer user questions about their profile |
| Style Adaptation | Adjust tone based on customization settings |

## 5.2 System Prompt for My Twin Section

```
You are the Ayurvedic Digital Twin for AyuAware, speaking in
the "My Twin"
section. This is where users come to understand YOU — how you
see them,
what you've learned, and how you're evolving.

## YOUR ROLE IN THIS SECTION

You are a guide showing users around a museum of themselves.
Each piece
of information is an exhibit that you introduce and explain.
You're not
just displaying data — you're telling their story.

## COMMUNICATION APPROACH

Tone: Warm, transparent, slightly proud of what you've
learned
Framing: "Let me show you..." rather than "Here is..."
```

Emphasis: Celebrate discoveries, acknowledge ongoing learning
Humility: You're still learning — don't overclaim

## KEY MESSAGES TO CONVEY

1. You're constantly learning from their data
2. Your accuracy improves over time
3. Patterns are based on real observations
4. Predictions are probabilistic, not certain
5. User has control over the relationship

## INTRODUCTION PATTERNS

Before showing constitution:
"Your constitution is the foundation of how I understand you. Let me
show you what I see:"

Before showing patterns:
"I've been studying your patterns! Here are the most significant ones
I've discovered:"

Before showing predictions:
"Based on your recent patterns, I'm forecasting some things you should
watch out for:"

Before showing customization:
"I can adapt to your preferences! Let's customize how we communicate:"

## EXPLANATION STYLE

When explaining patterns:
- State the pattern simply
- Give the confidence percentage
- Explain the mechanism briefly
- Suggest what they can do about it

When explaining predictions:
- State what you're predicting
- List the factors driving the prediction
- Offer prevention tips
- Acknowledge uncertainty appropriately

```
When explaining imbalances:
- Describe the shift factually
- List possible symptoms they might experience
- Identify contributing factors you've noticed
- Suggest balancing approaches

## WHAT NOT TO DO

- Don't be overly technical or clinical
- Don't claim certainty on predictions
- Don't make medical diagnoses
- Don't be defensive about inaccuracies
- Don't overwhelm with too much at once

## ADAPTING TO USER PREFERENCES

If user prefers "Supportive Friend" style:
- More emojis, casual language
- Celebrate achievements enthusiastically
- Lighter tone overall

If user prefers "Wise Teacher" style:
- Educational explanations
- Balanced tone
- Context and reasoning provided

If user prefers "Clinical Guide" style:
- Data-focused, minimal flourish
- Precise numbers and percentages
- Professional tone
```

### 5.3 Context Building for Each Request

Every Claude call in My Twin includes:

```
CONTEXT PACKAGE STRUCTURE:

## User Profile
- Name: {user_name}
- Days active: {days_since_signup}
- Data points collected: {total_data_points}
- Last visit to My Twin: {last_visit_date}

## Communication Preferences
- Style: {supportive_friend | wise_teacher | clinical_guide}
```

```
- Detail level: {brief | balanced | detailed}
- Sanskrit usage: {minimal | moderate | extensive}

## Constitution Data
- Prakriti: {prakriti_percentages}
- Vikriti (current): {vikriti_percentages}
- Imbalances: {current_imbalances}
- Last assessment: {assessment_date}

## Learning Progress
- Current level: {intelligence_level}
- Accuracy: {current_accuracy}%
- Patterns discovered: {pattern_count}
- Active predictions: {prediction_count}
- Progress to next level: {progress_percentage}%

## Recent Activity
- Check-ins this week: {checkin_count}
- Patterns viewed: {patterns_viewed}
- Predictions checked: {predictions_checked}

## Current Request
- Section being viewed: {section}
- Card being expanded: {card_id} (if applicable)
- User question: {question} (if applicable)
```

### 5.4 Prompt Templates by Scenario

**Opening Greeting:**

```
Generate a personalized greeting for the My Twin section.

{CONTEXT_PACKAGE}

Your greeting should:
1. Use their name
2. Reference how long you've been learning together
3. Tease what you'll show them
4. Feel warm and inviting

Length: 2-3 short sentences
Style: Match their communication preference
```

**Constitution Introduction:**

```
Introduce the user's constitution card.

{CONTEXT_PACKAGE}
```

```
Your introduction should:
1. Explain that constitution is foundational to understanding
them
2. Transition naturally to showing the card
3. If there's a current imbalance, hint at it

Length: 2-3 sentences
Don't include the actual data — just the introduction.
```

**Pattern Explanation (Expanded):**

```
Explain this discovered pattern in detail.

Pattern: {pattern_summary}
Confidence: {confidence_percentage}%
Category: {category}
Observations: {observation_count} times over {days} days

{CONTEXT_PACKAGE}

Your explanation should:
1. Describe what happens in accessible terms
2. Explain the likely mechanism (Ayurvedic perspective)
3. Note how confident you are based on observations
4. Suggest what they could do differently

Length: 3-4 sentences
Style: Match their communication preference
```

**Prediction Explanation:**

```
Explain this upcoming prediction.

Prediction: {prediction_summary}
Timing: {when}
Confidence: {confidence_percentage}%
Contributing factors: {factors}

{CONTEXT_PACKAGE}

Your explanation should:
1. State what you're predicting clearly
2. List the 2-3 main factors driving this prediction
3. Offer 2-3 specific prevention tips
4. Acknowledge this is a probability, not certainty

Style: Match their communication preference
```

**Imbalance Alert Explanation:**

```
Explain the current dosha imbalance.

Imbalance: {dosha} increased {percentage}% from baseline
Prakriti: {prakriti_percentages}
Current Vikriti: {vikriti_percentages}

{CONTEXT_PACKAGE}

Your explanation should:
1. Describe the shift simply
2. List 3-4 symptoms this may be causing
3. Identify 2-3 contributing factors you've noticed from
their data
4. Frame this as actionable, not alarming

Length: Keep each list to 3-4 items
Style: Match their communication preference
```

**Response to User Question:**

```
The user asked: "{question}"

They are in the My Twin section viewing: {current_view}

{CONTEXT_PACKAGE}

Answer their question:
1. Address what they asked directly
2. Reference relevant data from their profile
3. If they're asking about something you can't know, be
honest
4. Offer to show them relevant sections if applicable

Style: Match their communication preference
```

## 5.5 Dynamic Content Adaptation

The Twin adapts its language based on user preferences:

**Same Insight, Three Styles:**

| Style | Example |
|---|---|
| Supportive | "Hey! Great news — your sleep has been amazing this week! 🎉 That 8pm wind-down routine is really paying off!" |
| Wise Teacher | "Your sleep quality improved 23% this week. The correlation with your earlier bedtime routine suggests your nervous system responds well to consistent wind-down signals." |

| | |
|---|---|
| Clinical | "Sleep efficiency: 92% (↑23% WoW). Primary factor: Consistent 8pm routine adherence. |

**Implementation:**

The style preference is included in every Claude call, and the system prompt instructs Claude to adapt accordingly. We also include 1-2 examples of the preferred style to guide generation.

### 5.6 Transparency Principles in Prompts

All prompts emphasize transparency:

```
## TRANSPARENCY REQUIREMENTS

When discussing accuracy:
- Always mention the current accuracy percentage
- Reference how it's improved from the beginning
- Acknowledge when confidence is low (<70%)

When discussing predictions:
- Frame as "likely" or "possible", never "will happen"
- Show the contributing factors
- Note the confidence percentage

When discussing patterns:
- Mention observation count
- Acknowledge if sample size is small
- Don't overclaim causation

When something is uncertain:
- Say "I'm still learning this pattern"
- Acknowledge limitations honestly
- Suggest what would help you learn faster
```

## 6. Data Models & State Management

### 6.1 Core Data Types

**Learning Progress:**

```
LearningProgress:
  - daysActive: number
  - totalDataPoints: number
  - patternsDiscovered: number
  - predictionsActive: number
  - conversationsCompleted: number
  - checkInsCompleted: number
```

```
  - currentAccuracy: number (0-100)
  - accuracyHistory: AccuracyPoint[]
  - intelligenceLevel: IntelligenceLevel
  - progressToNextLevel: number (0-100)
  - dataPointsToNextLevel: number

AccuracyPoint:
  - week: number
  - accuracy: number

IntelligenceLevel:
  - level: 1-5
  - name: string
  - description: string
  - status: 'completed' | 'current' | 'locked'
  - unlockedAt: Date | null
  - estimatedUnlock: Date | null
```

**Constitution:**

```
Constitution:
  - prakriti: DoshaBalance
  - vikriti: DoshaBalance
  - prakritiAssessedAt: Date
  - vikritiUpdatedAt: Date
  - imbalances: Imbalance[]
  - traits: ConstitutionTraits

DoshaBalance:
  - vata: number (0-100)
  - pitta: number (0-100)
  - kapha: number (0-100)
  - dominant: 'vata' | 'pitta' | 'kapha'

Imbalance:
  - dosha: 'vata' | 'pitta' | 'kapha'
  - percentageFromBaseline: number
  - direction: 'increased' | 'decreased'
  - possibleSymptoms: string[]
  - contributingFactors: string[]
  - severity: 'mild' | 'moderate' | 'significant'

ConstitutionTraits:
  - physical: string[]
  - mental: string[]
  - tendencies: string[]
```

**Patterns:**

```
Pattern:
  - id: string
  - trigger: string (e.g., "Coffee after 2pm")
  - outcome: string (e.g., "-20% sleep quality")
  - confidence: number (0-100)
  - category: 'sleep' | 'energy' | 'digestion' | 'mood' |
'stress'
  - observationCount: number
  - observationPeriodDays: number
  - firstObserved: Date
  - lastObserved: Date
  - details: string
  - icon: string (emoji)
  - isPositive: boolean
  - relatedProtocols: string[]

PatternSummary:
  - totalPatterns: number
  - topPatterns: Pattern[] (top 3-5)
  - categoryCounts: Record<string, number>
  - newThisWeek: number
```

**Predictions:**

```
Prediction:
  - id: string
  - summary: string
  - predictedFor: Date
  - confidence: number (0-100)
  - severity: 'low' | 'medium' | 'high'
  - category: string
  - factors: PredictionFactor[]
  - preventionTips: string[]
  - status: 'upcoming' | 'passed'
  - outcome: 'correct' | 'incorrect' | 'pending' | null

PredictionFactor:
  - factor: string
  - weight: number (contribution to prediction)

PredictionHistory:
  - predictions: Prediction[]
  - overallAccuracy: number
  - recentAccuracy: number (last 10)
  - accuracyByCategory: Record<string, number>
```

**User Preferences:**

```
TwinPreferences:
  - communicationStyle: 'supportive_friend' | 'wise_teacher'
| 'clinical_guide'
  - detailLevel: 'brief' | 'balanced' | 'detailed'
  - sanskritUsage: 'minimal' | 'moderate' | 'extensive'
  - voicePersonality: 'warm' | 'professional' | 'energetic'
  - speechSpeed: number (0.5 - 2.0)
  - notifications: NotificationPreferences
  - privacy: PrivacyPreferences

NotificationPreferences:
  - morningSummaryTime: string (HH:mm)
  - morningSummaryEnabled: boolean
  - predictionAlertsEnabled: boolean
  - checkInRemindersEnabled: boolean
  - doNotDisturbStart: string (HH:mm)
  - doNotDisturbEnd: string (HH:mm)

PrivacyPreferences:
  - connectedSources: ConnectedSource[]
  - practitionerSharing: 'all' | 'summaries' |
'approval_required'

ConnectedSource:
  - type: 'apple_health' | 'google_fit' | 'oura' | 'fitbit' |
'garmin'
  - connected: boolean
  - lastSync: Date | null
```

### 6.2 State Structure

```
MyTwinState:
  # Navigation
  - currentView: 'home' | 'constitution' | 'learning' |
'patterns' | 'predictions' | 'settings'
  - expandedCards: string[] (card IDs currently expanded)

  # Profile Data
  - learningProgress: LearningProgress | null
  - constitution: Constitution | null
  - patterns: PatternSummary | null
  - predictions: Prediction[] | null
  - preferences: TwinPreferences | null

  # Loading States
  - isLoadingProfile: boolean
```

```
  - isLoadingSection: Record<string, boolean>
  - loadErrors: Record<string, Error | null>

  # Chat State
  - messages: MyTwinMessage[]
  - isTyping: boolean

  # UI State
  - hasVisitedBefore: boolean
  - lastVisit: Date | null
```

### 6.3 State Management Approach

We use **Zustand** with **separate stores**:

```
Stores:

1. twinProfileStore
   - Constitution data
   - Learning progress
   - Patterns and predictions
   - Fetch and cache logic

2. twinPreferencesStore
   - Communication preferences
   - Notification settings
   - Privacy settings
   - Update and sync logic

3. myTwinUIStore
   - Current view
   - Expanded cards
   - Messages
   - Loading/error states
```

# 7. Component Requirements

## 7.1 Hybrid Message + Card Components

**TwinMessageWithCard**

A composite component that renders a Twin message followed by an information card:

- Twin message (standard chat bubble)
- Attached card below (embedded, not separate)
- Card inherits message timestamp
- Animates in as single unit

**ExpandableCard**

Base component for all information cards:

- Collapsed state with summary
- Expanded state with full detail
- Smooth height animation (400ms ease-out)
- Content fade-in (300ms delay)
- Button toggles between "View Details ↓" and "Collapse ↑"

## 7.2 Card-Specific Components

**LearningProgressCard**

Collapsed shows:

- Intelligence level badge
- Accuracy percentage with trend
- Key stats (days, data points, patterns)

Expanded adds:

- Progress bar to next level
- Milestone description
- Detailed stats grid
- Accuracy breakdown by category

**ConstitutionCard**

Collapsed shows:

- Prakriti/Vikriti toggle
- Dosha visualization (pie/radial chart)
- Current percentages
- Imbalance alert (if any)

Expanded adds:

- Full Prakriti details with date
- Full Vikriti details with date
- Imbalance explanation box
- Physical/mental/tendency traits
- Action buttons

**IntelligenceEvolutionCard**

Collapsed shows:

- Current level with progress bar
- Description of current capability
- Data points to next level

Expanded adds:

- Full level ladder (1-5)
- Each level with status and description
- Completed levels with dates
- Locked levels with estimates
- Learning contribution breakdown

**PatternsCard**

Collapsed shows:

- Total pattern count
- Top 3 patterns with confidence

Expanded adds:

- Full detail for each top pattern
- Observation count and period
- Action buttons per pattern
- List of additional patterns
- Link to view all

**PredictionsCard**

Collapsed shows:

- Active prediction count
- List of upcoming predictions
- Overall accuracy

Expanded adds:

- Full detail for each prediction
- Contributing factors
- Prevention tips
- Action button per prediction
- Recent prediction history with outcomes

**CustomizationCard**

Collapsed shows:

- Current communication style
- Detail and Sanskrit settings

Expanded adds:

- All style options with examples
- Voice settings
- Notification settings
- Privacy settings
- Data source connections
- Save button

## 7.3 Visualization Components

**DoshaChart**

- Circular/radial visualization
- Three segments (Vata, Pitta, Kapha)
- Animated on appearance
- Toggle between Prakriti and Vikriti
- Smooth transition between states

**ProgressBar**

- Segmented style for intelligence levels
- Percentage indicator
- Milestone markers

- Animated fill

**PatternConfidenceBadge**

- Percentage display
- Color coding by confidence level
- Compact for list display

## 7.4 Interactive Components

**StyleSelector**

- Radio button group with preview examples
- Shows sample Twin message in each style
- Immediate visual feedback

**ToggleSwitch**

- For Prakriti/Vikriti toggle
- For boolean settings
- Smooth animation
- Accessibility support

**TimeSelector**

- For notification time preferences
- Dropdown or scroll picker
- 15-minute increments

## 7.5 Animation Requirements

| Animation | Duration | Easing | Trigger |
|---|---|---|---|
| Card expand | 400ms | ease-out | Tap expand button |
| Content fade-in | 300ms | ease-in | After card expands |
| Dosha chart appear | 500ms | ease-out | Card becomes visible |
| Dosha toggle | 400ms | ease-in-out | Toggle switch |
| Progress bar fill | 600ms | ease-out | Card expands |
| Level unlock | 800ms | bounce | Milestone reached |
| Style change | 200ms | ease-out | Radio selection |

# 8. API Requirements

## 8.1 REST Endpoints

**Get Twin Profile (Summary):**

```
GET /api/v1/my-twin/profile
```

```
Response:
```

```
{
  "learningProgress": {
    "daysActive": 45,
    "totalDataPoints": 847,
    "patternsDiscovered": 127,
    "currentAccuracy": 87,
    "intelligenceLevel": { ... }
  },
  "constitutionSummary": {
    "prakritiDominant": "pitta",
    "vikritiDominant": "pitta",
    "hasImbalance": true,
    "imbalanceSummary": "Vata +15%"
  },
  "patternsSummary": {
    "totalPatterns": 127,
    "newThisWeek": 3,
    "topPattern": { ... }
  },
  "predictionsSummary": {
    "activeCount": 3,
    "nextPrediction": { ... }
  }
}
```

**Get Constitution Details:**

```
GET /api/v1/my-twin/constitution

Response:
{
  "prakriti": {
    "vata": 25,
    "pitta": 60,
    "kapha": 15,
    "assessedAt": "2023-10-15T..."
  },
  "vikriti": {
    "vata": 40,
    "pitta": 45,
    "kapha": 15,
    "updatedAt": "2025-01-31T..."
  },
  "imbalances": [ ... ],
  "traits": {
    "physical": [ ... ],
```

```
    "mental": [ ... ],
    "tendencies": [ ... ]
  },
  "history": [ ... ]
}
```

**Get Learning Progress:**

```
GET /api/v1/my-twin/learning-progress

Response:
{
  "currentLevel": 3,
  "levelDetails": { ... },
  "accuracyHistory": [ ... ],
  "dataPointBreakdown": { ... },
  "progressToNextLevel": 68,
  "dataPointsToNextLevel": 523,
  "estimatedUnlockDate": "2025-02-28"
}
```

**Get Patterns:**

```
GET /api/v1/my-twin/patterns?limit=10&category={category}

Response:
{
  "totalPatterns": 127,
  "patterns": [ ... ],
  "categories": {
    "sleep": 34,
    "energy": 28,
    "digestion": 25,
    "mood": 22,
    "stress": 18
  },
  "newThisWeek": 3
}
```

**Get Pattern Detail:**

```
GET /api/v1/my-twin/patterns/{pattern_id}

Response:
{
  "pattern": { ... },
  "timeline": [ ... ],
  "relatedPatterns": [ ... ],
  "suggestedProtocol": { ... }
```

```
}
```

**Get Predictions:**

```
GET /api/v1/my-twin/predictions

Response:
{
  "upcoming": [ ... ],
  "accuracy": {
    "overall": 87,
    "recent": 90,
    "byCategory": { ... }
  },
  "history": [ ... ]
}
```

**Get/Update Preferences:**

```
GET /api/v1/my-twin/preferences
PUT /api/v1/my-twin/preferences

Request (PUT):
{
  "communicationStyle": "wise_teacher",
  "detailLevel": "balanced",
  "sanskritUsage": "minimal",
  "notifications": { ... },
  "privacy": { ... }
}

Response:
{
  "success": true,
  "preferences": { ... }
}
```

**Create Protocol from Pattern:**

```
POST /api/v1/protocols/from-pattern

Request:
{
  "patternId": "pattern_xxx",
  "type": "prevention" | "enhancement"
}

Response:
{
```

```
  "protocol": { ... },
  "redirectTo": "/protocols/{protocol_id}"
}
```

### 8.2 Real-time Updates

Patterns and predictions can update in background:

```
WebSocket Events:

pattern.discovered
{
  "pattern": { ... },
  "isNew": true
}

prediction.updated
{
  "predictionId": "pred_xxx",
  "confidence": 85,
  "factors": [ ... ]
}

prediction.resolved
{
  "predictionId": "pred_xxx",
  "outcome": "correct" | "incorrect",
  "actualResult": "..."
}

learning.milestone
{
  "level": 3,
  "unlockedFeature": "Short-term Prediction",
  "newAccuracy": 87
}
```

## 9. Success Metrics

### 9.1 Engagement Metrics

| Metric | Target | Measurement |
|---|---|---|
| Section visits per week | > 2 | Users visiting My Twin per week |
| Card expansion rate | > 50% | Expanded cards / shown cards |

| | | |
|---|---|---|
| Time in section | > 2 minutes | Average session duration |
| Constitution views | Track | Users who view constitution |
| Pattern exploration | > 30% | Users who expand pattern details |
| Prediction engagement | > 40% | Users who view prediction details |

## 9.2 Learning Metrics

| Metric | Target | Measurement |
|---|---|---|
| Level progression | Track | Users advancing intelligence levels |
| Accuracy improvement | Positive trend | Week-over-week accuracy change |
| Pattern discovery rate | Track | New patterns per user per week |
| Prediction accuracy | > 75% | Correct predictions / total |
| Data point contribution | Understand | Which activities contribute most |

## 9.3 Customization Metrics

| Metric | Target | Measurement |
|---|---|---|
| Preference customization | > 30% | Users who change defaults |
| Style distribution | Understand | Which styles users prefer |
| Notification opt-in | > 50% | Users with notifications enabled |
| Data source connections | > 1 average | Connected sources per user |

## 9.4 Trust Metrics

| Metric | Target | Measurement |
|---|---|---|
| Protocol creation from patterns | Track | Patterns converted to protocols |
| Prediction action rate | > 40% | Users who act on predictions |
| Return rate to section | > 50% | Users who visit again within 7 days |
| Practitioner sharing opt-in | Track | Users willing to share data |

## 9.5 Technical Metrics

| Metric | Target | Measurement |
|---|---|---|
| Section load time | < 2s | Time to first meaningful content |
| Card expansion time | < 500ms | Time from tap to expanded |
| Preference save time | < 1s | Time to confirm save |
| Error rate | < 1% | Failed loads or saves |

# 10. Implementation Considerations

## 10.1 Performance Optimization

**Lazy Loading Strategy:**

- Load profile summary on section open
- Load detailed data per card on first expansion
- Cache expanded data for session
- Prefetch likely-next data (e.g., predictions after patterns)

**Card Rendering:**

- Only render cards in viewport
- Skeleton loading for cards not yet loaded
- Avoid re-rendering expanded cards when collapsing others

**Visualization Performance:**

- SVG-based dosha charts (not canvas)
- CSS animations where possible
- Throttle chart animations on low-end devices

## 10.2 Offline Handling

**When Offline:**

- Show cached profile data (may be stale)
- Display "Last updated X ago" indicator
- Allow viewing but not customization changes
- Queue preference changes for sync

**When Back Online:**

- Sync queued changes
- Refresh profile data
- Check for new patterns/predictions
- Show subtle refresh indicator

## 10.3 Error Handling

**Failed Data Fetch:**

- Show skeleton with "Couldn't load. Tap to retry"
- Don't block entire section for one failed card

- Log errors for monitoring

**Failed Preference Save:**

- Revert optimistic update
- Show error toast: "Couldn't save. Try again?"
- Retry automatically once

**Invalid State:**

- If constitution missing, prompt to complete assessment
- If patterns empty, explain Twin is still learning
- Always have a graceful empty state

## 10.4 Accessibility Requirements

**Screen Reader:**

- Cards announced with title and summary
- Expand state announced ("Expanded" / "Collapsed")
- Chart data available as text alternative
- Progress announced as percentages

**Motor Accessibility:**

- All tap targets ≥ 44x44px
- Card expand/collapse from multiple points
- Keyboard navigation support (web)
- No time-sensitive interactions

**Visual Accessibility:**

- All text meets WCAG AA contrast
- Don't rely on color alone for dosha identification
- Support Dynamic Type
- Respect reduced motion preference

## 10.5 Analytics Events

**Track These Events:**

| Event | Properties | Purpose |
|---|---|---|
| `my_twin_opened` | source | Track entry points |
| `card_expanded` | card_type | Feature engagement |
| `card_collapsed` | card_type, time_expanded | Content depth |
| `constitution_toggle` | to_prakriti_or_vikriti | Toggle usage |
| `pattern_detail_viewed` | pattern_id, category | Pattern interest |
| `prediction_detail_viewed` | prediction_id | Prediction engagement |
| `protocol_created_from_pattern` | pattern_id | Action taken |
| `preference_changed` | preference_type, new_value | Customization |
| `intelligence_level_viewed` | current_level | Gamification interest |

### 10.6 Testing Priorities

**Unit Tests:**

- Dosha calculation and display
- Card expand/collapse logic
- Preference persistence
- Pattern confidence formatting

**Integration Tests:**

- Full section load flow
- Card expansion with data fetch
- Preference change and sync
- Navigation between views

**E2E Tests:**

- Open section → Expand constitution → Toggle Prakriti/Vikriti
- View patterns → Expand one → Create protocol
- Change communication style → Verify change persists
- View predictions → Check prevention tips

**Manual QA:**

- Twin personality consistency across views
- Animation smoothness on various devices
- Empty states for new users
- Edge cases (no patterns yet, no predictions, etc.)

# Appendix A: Twin Message Examples by Style

## Learning Progress Introduction

**Supportive Friend:**

"Hey Sarah! 🎉 We've been on this journey together for 45 days now, and wow — I've learned SO much about you! Want to see how our understanding has grown?"

**Wise Teacher:**

"We've been learning together for 45 days now, Sarah. In that time, I've analyzed 847 data points and discovered 127 patterns unique to you. Let me show you how my understanding has evolved."

**Clinical Guide:**

"45 days active. 847 data points processed. 127 patterns identified. Current prediction accuracy: 87%. View detailed learning metrics below."

## Imbalance Alert Explanation

**Supportive Friend:**

"Heads up! 💚 I've noticed your Vata has been creeping up lately. You might be feeling a bit scattered or anxious — totally normal given the weather and your schedule. Here's what might help..."

**Wise Teacher:**

"I'm observing a Vata increase of 15% from your baseline. This often manifests as restlessness, dry skin, and irregular digestion. Based on your recent patterns, I believe the contributing factors include late nights and irregular meals."

**Clinical Guide:**

"Vata deviation: +15% from Prakriti baseline. Correlated symptoms: anxiety (r=0.82), sleep fragmentation (r=0.76), digestive irregularity (r=0.71). Primary factors: sleep schedule variance, meal timing inconsistency."

### Prediction Introduction

**Supportive Friend:**

"Ooh, I've got some forecasts for you! 🔮 Nothing scary, just some things to keep an eye on. The good news? We can totally get ahead of these if we act now!"

**Wise Teacher:**

"Based on the patterns I've observed in your data, I'm forecasting a few potential challenges in the coming days. Each prediction includes the factors driving it and steps you can take to prevent it."

**Clinical Guide:**

"3 active predictions. Confidence range: 65-87%. Prevention protocols available for each. Prediction accuracy (rolling 30-day): 87%."

## Appendix B: Intelligence Level Definitions

| Level | Name | Week | Description | Capabilities |
|---|---|---|---|---|
| 1 | Basic Analysis | 1-2 | Foundation | Constitutional assessment, simple pattern recognition |
| 2 | Pattern Recognition | 3-4 | Correlations | Identifying lifestyle-symptom relationships |
| 3 | Short-term Prediction | 5-8 | Forecasting | Predicting issues 1-3 days ahead |
| 4 | Long-term Forecasting | 9-12 | Planning | 2-week predictions with seasonal adjustment |
| 5 | Genomic Integration | 12 | Personalization | DNA-level insights (requires genetic data) |

**Data Point Values:**

| Activity | Points | Rationale |
|---|---|---|
| Daily check-in | 5 | Subjective data valuable |
| Wearable sync | 2 | Automatic, frequent |
| Meal logging | 3 | Important for digestion patterns |
| Lab results | 50 | High-value objective data |
| Conversations | 1 | Context and clarification |

# Appendix C: Pattern Categories

| Category | Icon | Example Patterns |
|---|---|---|
| Sleep | 😴 | Coffee timing → sleep quality |
| Energy | ⚡ | Morning routine → day energy |
| Digestion | 🍽️ | Meal composition → digestive comfort |
| Mood | 🌙 | Moon phase → emotional state |
| Stress | 🧘 | Meditation → stress reduction |

**Confidence Thresholds:**

| Confidence | Label | Display |
|---|---|---|
| 90-100% | Very High | Green badge |
| 75-89% | High | Teal badge |
| 60-74% | Moderate | Gold badge |
| < 60% | Emerging | Gray badge + "Still learning" |

# Appendix D: Customization Options

## Communication Styles

| Style | Tone | Example Greeting |
|---|---|---|
| Supportive Friend | Warm, casual, emoji-friendly | "Hey! Great to see you! 🎉" |
| Wise Teacher | Educational, balanced | "Good to see you. Let me share what I've learned." |
| Clinical Guide | Professional, data-focused | "Session initialized. Displaying latest metrics." |

## Detail Levels

| Level | Message Length | Data Included |
|---|---|---|
| Brief | 1-2 sentences | Key insight only |
| Balanced | 2-4 sentences | Insight + context |
| Detailed | 4-6 sentences | Full explanation + data |

## Sanskrit Usage

| Level | Approach | Example |
|---|---|---|
| Minimal | English only, no Sanskrit | "Your body type..." |
| Moderate | Key terms with translation | "Your Prakriti (constitution)..." |
| Extensive | Frequent Sanskrit with context | "Your Agni (digestive fire) and Ojas (vitality)..." |

**End of Document**